

Midterm Exam — Practice Problems

References: Munson, B., Rothmayer, A., Okiishi, T., & Huebsch, W. (2013). *Fundamentals of fluid mechanics* (7th ed.). John Wiley & Sons [M]; Tannehill, J. C., Anderson, D. A., & Pletcher, R. H. (1997). *Computational fluid mechanics and heat transfer* (2nd ed.). Taylor & Francis [TAP].
Course repository: <https://github.com/wbandabarragan/computational-fluid-dynamics>

The midterm exam will have 2 sections:

1. **Multiple-choice and questions:** short problem and concept questions, of the kind listed in Parts A and B below (similar to the quizzes).
2. **Code and problems:** questions on the code and on the problems of the exam notebook: <https://github.com/wbandabarragan/computational-fluid-dynamics/blob/main/exams/midterm-exam.ipynb>.

How to work through it

- Work in SI throughout. If the textbooks pose a problem in British Gravitational units, convert the data first and answer in SI.
- For every exercise, ask yourself what the code would look like: the inputs, the function that does the work, the figure, and the check.
- Estimate the order of magnitude before you compute, and check it afterwards. Attempt each problem yourself before looking anything up. Submission is not needed.

Part A — Concepts

Unit 1 — Fluid properties and fluid statics

- L1.** Definition of a fluid; the continuum hypothesis; the ideal gas law; Newton's law of viscosity; Newtonian and non-Newtonian fluids; the Reynolds number.
- L2.** Compressibility and the bulk modulus; the speed of sound; vapour pressure and cavitation; surface tension, contact angle and capillary rise.
- L3.** The hydrostatic equation; incompressible and compressible fluids; the isothermal atmosphere and the scale height; the standard atmosphere; absolute and gauge pressure.
- L4.** U-tube, differential and inclined manometers; force on a plane surface, the pressure prism and the centre of pressure; the two components of the force on a curved surface.
- L5.** Archimedes' principle and the centre of buoyancy; flotation; stability, the metacentre, and the relative positions of the centres of gravity and of buoyancy.

Unit 2 — The Bernoulli equation and fluid kinematics

- L6.** Newton's law along and normal to a streamline; the Bernoulli equation and its four restrictions; static, dynamic, stagnation and total pressure; the free vortex.
- L7.** Free jets; the Pitot-static tube; the Venturi, nozzle and orifice meters; continuity in confined flows; cavitation; the limit $Ma \lesssim 0.3$.
- L8.** Eulerian and Lagrangian descriptions; streamlines, pathlines and streaklines; the material derivative, local and convective acceleration; the Reynolds transport theorem.

Unit 3 — Foundations of CFD and the governing equations

- L9.** Equilibrium and marching problems; characteristics and the discriminant; canonical forms; the 3 PDE types; domains of dependence; well-posedness and boundary data.
- L10.** Conservation of mass, momentum and energy; the continuity equation; the Euler equations and the equation of state; the viscous stress tensor; the Navier–Stokes equations.

Part B — Short exercises

Unit 1 — Lectures 1–5. Check what you have already done:

[quiz-practice/quiz1_practice.pdf](#)
[notebooks/nb01_standard_atmosphere.ipynb](#)
[notebooks/nb02_hydrostatic_forces.ipynb](#)
[homework/homework1.ipynb](#)

New practice problems:

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|---|--|
| 1. M 1.44 — ideal gas law. | 5. M 2.48 — manometry, unknown fluid. |
| 2. M 1.79 — piston, viscous force. | 6. M 2.77 — force, plane gate. |
| 3. M 1.94 — bulk modulus. | 7. M 2.116 — force, curved gate. |
| 4. M 2.23 — compressible atmosphere. | 8. M 2.148 — buoyancy, iceberg. |

Unit 2 — Lectures 6–8. Check what you have already done:

[quiz-practice/quiz2_practice.pdf](#)
[notebooks/nb03_streamlines_stagnation_points.ipynb](#)
[notebooks/nb04_bernoulli_venturi_effects.ipynb](#)

New practice problems:

- | | |
|--|---|
| 1. M 3.8 — gradient past cylinder. | 5. M 4.13 — equation of streamlines. |
| 2. M 3.27 — wind over roof. | 6. M 4.32 — acceleration along streamline. |
| 3. M 3.74 — open-circuit wind tunnel. | 7. M 4.57 — material derivative, pressure. |
| 4. M 3.117 — cavitation, Venturi meter. | 8. M 4.71 — control volume, channel. |

Unit 3 — Lectures 9–10. Check what you have already done:

[exams/midterm-exam.ipynb](#)

Lecture 10 will be examined by concepts only. For Lecture 9, check these 4 problems from Chapter 2 of **TAP**: python can help you.

1. **TAP 2.9** — classification, canonical form.
2. **TAP 2.12** — character of a system.
3. **TAP 2.20** — wave equation, d'Alembert.
4. **TAP 2.21** — Laplace by separation.

If you want to solve more: hydrostatic forces, M 2.78 to 2.94 on plane surfaces and M 2.114 to 2.128 on curved ones; free jets and siphons, M 3.58 to 3.62; flowmeters, M 3.116 to 3.120; acceleration fields, M 4.32 to 4.47; canonical forms, TAP 2.16 to 2.19.

Important dates: Midterm Exam

- **Midterm Written Exam:** Wednesday 7 October 2026 (at 5pm, in class)
- **Notebook Submission Deadline:** Thursday 8 October 2026 (by 9pm)
- **Credits:** 25 points